

EXPLORATORY DATA ANALYSIS (EDA) ON E-COMMERCE DATASET

Project Report

Submitted By:

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Internship Program:

Data Analytics Internship

Organization:

DecodeLabs

Project Title:

Project 2 – Exploratory Data Analysis (EDA)

DECLARATION

I hereby declare that the project titled "Exploratory Data Analysis (EDA) on E-Commerce Dataset" is my original work completed as part of the DecodeLabs Data Analytics Internship. The work presented in this report has been carried out by me and has not been submitted elsewhere for any academic or professional purpose.

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Exploratory Data Analysis (EDA) on E-Commerce Dataset

1. Introduction

Exploratory Data Analysis (EDA) is a process used to understand the structure, patterns, trends, and relationships present in a dataset. The objective of this project is to analyze an e-commerce dataset and extract meaningful insights that can support business decision-making. The analysis includes descriptive statistics, distribution analysis, outlier detection, and correlation analysis.

2. Dataset Description

The dataset contains information related to customer orders in an e-commerce system.

Dataset Summary

- Total Records: 1200
- Total Columns: 14
- Time Period: January 2023 to June 2025

Main Attributes

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- ShippingAddress
- PaymentMethod
- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource
- TotalPrice

3. Data Cleaning

The dataset used for this analysis was cleaned in Project 1.

The following steps were performed:

- Checked for missing values
- Removed duplicate records
- Verified date formats
- Standardized data structure
- Validated data consistency

After cleaning, the dataset contained no missing values and was ready for analysis.

4. Statistical Analysis

Descriptive statistics were calculated to understand the dataset.

Key Statistics

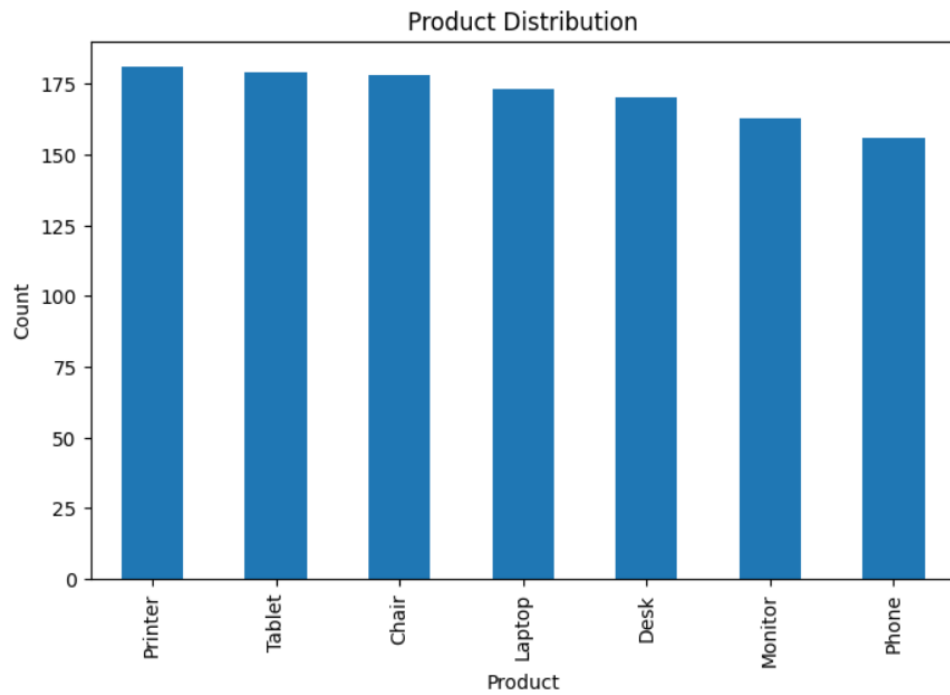
Metric	Quantity	UnitPrice	ItemsInCart	TotalPrice
Mean	2.95	356.41	5.49	1053.97
Median	3	364.21	5	823.62
Maximum	5	699.93	10	3456.40

The average order value was ₹1053.97, while the median order value was ₹823.62.

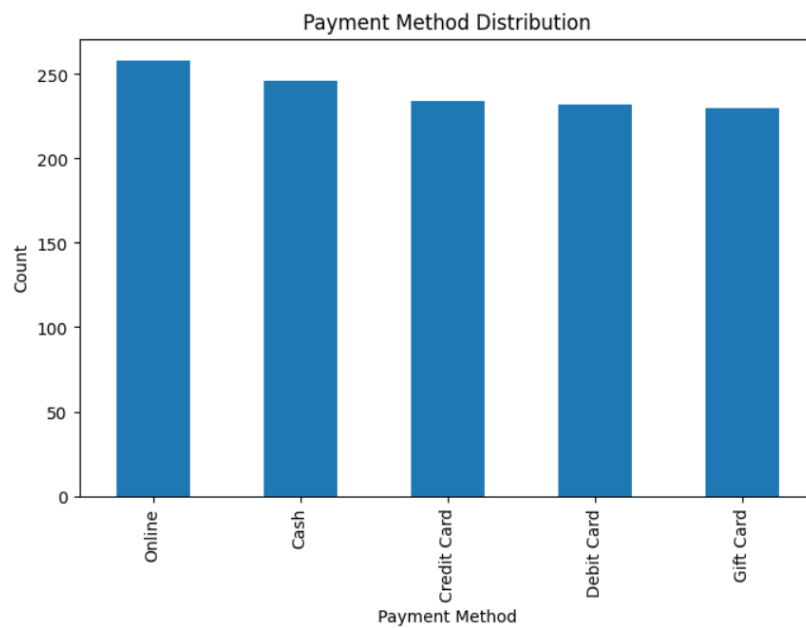
5. Visualizations

Several visualizations were created to understand the dataset:

1. Product Distribution Bar Chart



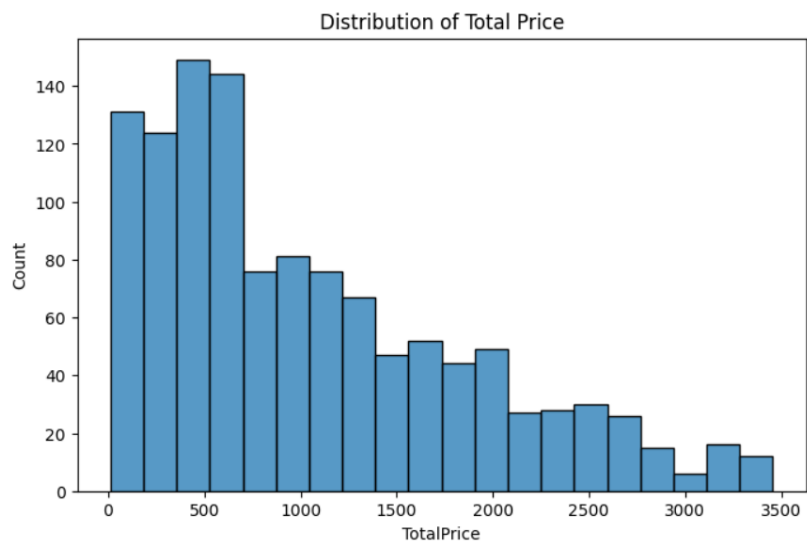
2. Payment Method Distribution Chart



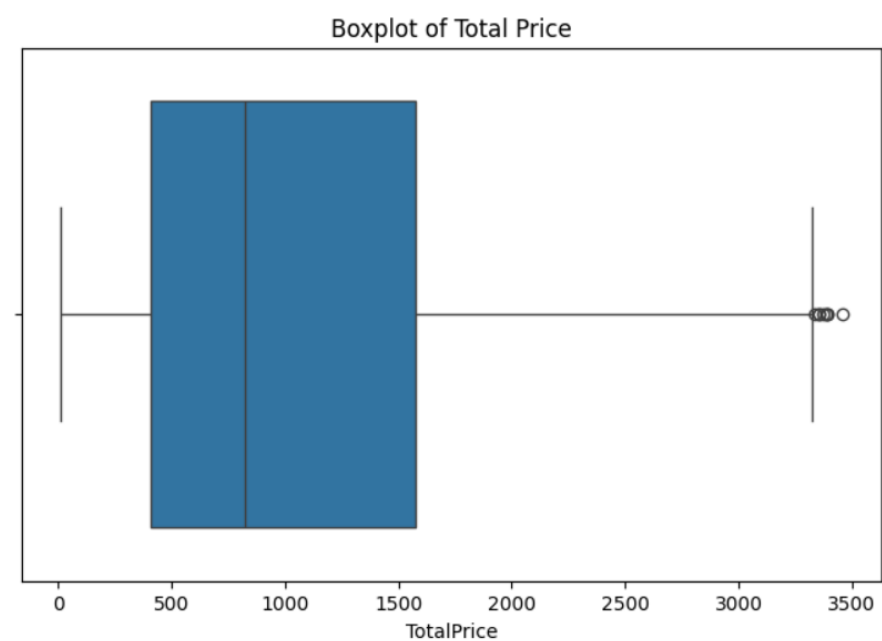
3. Order Status Distribution Chart



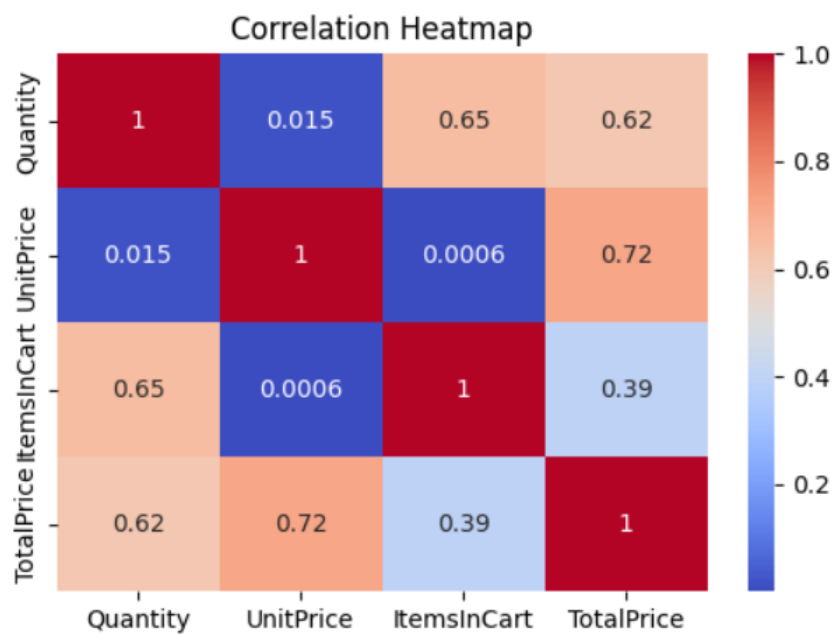
4. Total Price Histogram



5. Total Price Boxplot



6. Correlation Heatmap



6. Outlier Analysis

A boxplot was used to identify outliers in the TotalPrice column.

Observation

A few high-value orders were detected beyond the upper whisker of the boxplot. These observations represent unusually expensive transactions compared to the majority of orders.

Such outliers may indicate:

- Premium purchases
- Bulk orders
- Special customer transactions

Therefore, these records were retained for business analysis.

7. Correlation Analysis

Correlation analysis was performed on numeric variables.

Key Correlations

Variables	Correlation
UnitPrice ↔ TotalPrice	0.72
Quantity ↔ ItemsInCart	0.65
Quantity ↔ TotalPrice	0.62
ItemsInCart ↔ TotalPrice	0.39

Interpretation

- UnitPrice and TotalPrice have a strong positive relationship.
- Customers purchasing more items generally generate higher order values.
- Quantity and UnitPrice show very weak correlation, indicating that product price does not strongly influence purchase quantity.

8. Key Findings

1. The average order value is ₹1053.97.
2. The median order value is ₹823.62.
3. Printer is the most frequently ordered product.
4. Online Payment is the most preferred payment method.
5. The dataset contains a few high-value outlier orders.
6. UnitPrice and TotalPrice have a strong positive correlation.
7. Customers with more items in their carts tend to purchase larger quantities.

9. Recommendations

- Focus marketing efforts on high-demand products.
- Encourage digital payment methods through rewards and cashback offers.
- Investigate high-value transactions for potential premium customer opportunities.
- Use customer purchasing patterns for targeted promotions and recommendations.

10. Conclusion

The exploratory data analysis successfully identified important patterns, trends, and relationships within the e-commerce dataset. The analysis revealed customer purchasing behavior, product demand, payment preferences, and revenue drivers. Correlation and outlier analysis provided additional business insights that can support future decision-making. Overall, the dataset demonstrated valuable opportunities for improving sales strategies and customer engagement.